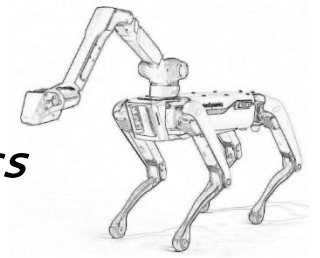


EE 587 Introduction to Robotics

Homework #3

(Given: Nov 26, Due: Dec 13 23:55)



General Info

- The homeworks are for individual study and are for gaining a better understanding of the course topics. Submit a homework that is your own effort.
- All explanations, results, figures and screenshots should be included in a pdf document. Show code fragments etc. for programming questions. For theoretical questions, you may use your handwriting on a tablet (or scanned segments) that are attached to the document at appropriate places. Your handwriting should be clear and legible in this case.

Questions

1. Suppose that $a = (1, -1, 2)$ and that $R = R_{x, 90}$. Show by direct calculation that you have $RS(a)R^T = S(Ra)$.
2. Given the Euler angle transformation $R = R_{z, \varphi} R_{y, \theta} R_{z, \phi}$, show that

$$\frac{d}{dt} R = S(\omega) R$$

Where $\omega = \{c_\psi s_\theta \dot{\phi} - s_\psi \dot{\theta}\}i + \{s_\psi s_\theta \dot{\phi} + c_\psi \dot{\theta}\}j + \{\dot{\psi} + c_\theta \dot{\phi}\}k$
 (the components of i, j and k are respectively called *nutation*, *spin* and *precession*).

3. Let $R_{k, \theta}$ be the rotation matrix corresponding to a rotation around an axis defined by the vector $k = (k_1, k_2, k_3)$. Prove using the Rodrigues' Formula that we have

$$\frac{d}{d\theta} R_{k, \theta} = S(k) R_{k, \theta}$$

4. Two frames $\{0\}$ and $\{1\}$ are related by the Homogeneous transformation

$$H = \begin{bmatrix} 0 & -1 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

A particle has velocity $v_1(t) = (3, 1, 0)$ relative to frame $\{1\}$. Calculate the particle velocity relative to frame $\{0\}$?

Programming:

5. In this question, you will implement a simple “Robot controller” by using the *forward kinematics* and *inverse velocity kinematics* of a planar RR manipulator. We know all the kinematics and velocity kinematics equations of this planar robot from lecture notes. You can use either Matlab or Python to implement this question.
 - a. Write code to “simulate” a Planar RR robot with link lengths $l_1 = 3$ and $l_2 = 2$ units. Define the angular ranges for the joints as $\theta_1 = \{-90^\circ, 90^\circ\}$ and $\theta_2 = \{-170^\circ, 170^\circ\}$. For any given pair of joint variables, you should be able to plot the robot on a plane (different colors for each link? Small circles for the joints? A fat point for the end effector?)
 - b. Define two points in the robot workspace: The start point p_s and goal point p_g in the robot work-space. (You can later experiment with different start and end points)
 - c. We would like to implement a discrete-time *proportional position controller* in the work-space that would take the robot from the initial point to the goal point. For this, you will periodically (at each time step): (1) calculate the position error, (2) define a required end-effector velocity vector (your controller), (3) transform this into required joint velocities, (4) execute the joint motion to determine the new end-effector position, (5) repeat the cycle.
 - d. Experiment with different sampling periods and controller Proportional Gains.
 - e. Present your results in the report as “time snapshots” of the robot configuration and/or end effector positions. Also see (g) below.
 - f. You should be able to detect and raise an alarm if a singularity is encountered during the control of the robot.
 - g. Have a video recording (screen capture) of the execution of your controller for some example cases. Upload to YouTube as “unlisted” videos and share the link in your homework report.
6. Review the P. Corke book and *Robotics Toolbox for Python* documentation to list (with short explanations) fundamental functions that calculate the *forward and inverse velocity kinematics* of robot arms as well as the *inverse position kinematics*. You may illustrate some examples that use these functions.

Good luck.